

Code:

```
#include <stdio.h>

void sort(int a[], int n, int flag) {
    int temp;
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (flag == 1) {
                // Ascending order
                if (a[i] > a[j]) {
                    temp = a[i];
                    a[i] = a[j];
                    a[j] = temp;
                }
            } else {
                // Descending order
                if (a[i] < a[j]) {
                    temp = a[i];
                    a[i] = a[j];
                    a[j] = temp;
                }
            }
        }
    }
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while (t--) {
        int n;
        if (scanf("%d", &n) != 1) break;

        int a[n], b[n];
        for (int i = 0; i < n; i++) scanf("%d", &a[i]);
        for (int i = 0; i < n; i++) scanf("%d", &b[i]);

        sort(a, n, 1); // Sort A ascending
        sort(b, n, 0); // Sort B descending
    }
}
```

```
long long min_sum = 0;
for (int i = 0; i < n; i++) {
    min_sum += (long long)a[i] * b[i];
}

printf("%lld\n", min_sum);
}

return 0;
}
```

Input:

```
3
4
2 5 3 7
8 5 7 3
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
```

Output:

```
83
247
451
```

